Highlights

The geometry of 16,369 breaking waves, re-decided in exact arithmetic from the published table, with the sea-state stratification the dataset allows*

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- Every number a dataset paper prints is re-derived from its own table, exactly.
- The three quadratic laws and the 4% self-similar tail reproduce to the printed digit.
- Two printed quantities are shown to be other columns of the public table.
- Per record, gL_b/c_b^2 spans a factor 1.66 and ranks with H_s at $\rho = -0.493$.
- A printed digit is read as a box; the verdict names which column reproduces it.

The geometry of 16,369 breaking waves, re-decided in exact arithmetic from the published table, with the sea-state stratification the dataset allows

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Abstract

A dataset paper prints a few dozen numbers that summarise a table: counts, ranges, fitted coefficients, correlation coefficients, the size of a tail. Guimarães et al. (2026a) measured the geometry of 16,369 actively breaking waves by stereo video on the Black Sea and published the per-event table and the analysis scripts beside the paper (Guimarães et al., 2026b). This paper reads every number the paper prints against that table in exact rational arithmetic, with the definitions the scripts use: each literal of the table is the rational its digits denote, a fit through the origin is a ratio of two integer sums, a correlation coefficient has an exact square, a quantile is a value that occurs in the table, and a printed digit is a box that the exact value either rounds into or does not. Of 25 numbers in the text, 13 are the rounding of the exact value, 1 its truncation and 1 its integer parts; 2 are coefficients of a randomised fitter that no exact procedure can reproduce; 8 are not the table's under the scripts' definitions. Of 140 cells of the paper's Table 1, 128 are the rounding. The three quadratic laws $L_b = 0.83\,c_b^2/g$, $L_{D81} = 0.66\,c_b^2/g$ and $\Delta z_b = 0.20\,c_b^2/g$ reproduce to the printed digit, and so do the $4 \pm 2\%$ self-similar tail and the fit on it. Two printed quantities are shown to be other columns of the public table: the printed range of the inclination θ is, to the integer part, the range of $\tan^{-1}(\Delta z_b/e_B)$, while the column named θ runs 1.38° to 42.97°; and the printed significant wave height of Table 1 is the column the table names `sv_fp2`, not the column it names `Hs`. Duncan's

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laboratory geometry is decided event by event, and the stratification by sea state that the paper names as beyond its scope is done per record: the coefficient of L_b on c_b^2/g runs from 0.665 to 1.106 across the twenty records and ranks with the record's H_s at Spearman $\rho = -0.493$, with wave age at -0.026. The novelty claimed is the decision grammar — a verdict per printed number, with the quantity that justifies it — and the public record that any reader can re-run without the authors' code.

Keywords: breaking waves, whitecaps, stereo video, self-similarity, exact arithmetic, verified computation, reproducibility

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1. Introduction

The geometry of a breaking wave — how long the whitecap is, how much area it covers, how steep its face, how fast it travels — has been measured mostly in tanks. Duncan (1981) towed a hydrofoil and found a breaking region of fixed shape: a face inclined between 10° and 14.7° and a cross-sectional area equal to 0.11 ± 0.01 of the square of its length. Phillips (1985) built the spectral framework of whitecap statistics on constants of that kind, and the framework is used today to estimate whitecap coverage, air entrainment, momentum flux and dissipation (Gemmrich et al., 2008; Banner et al., 2014b; Sutherland and Melville, 2013). Field observations of the geometry itself are few (Kleiss and Melville, 2011). Guimarães et al. (2026a) published the largest set so far: 16,369 actively breaking events filmed in stereo from a platform in the Black Sea over 20 records, with the area, length, vertical extent, duration and speed of each, and found that the field geometry does not conform to a single self-similar shape, that the breaking speed is a weak predictor of the geometry while the geometric properties scale together, and that only a $4 \pm 2\%$ tail of events is self-similar. Beside the paper they published the per-event table and the scripts that drew its figures (Guimarães et al., 2026b).

The findings of such a paper are statements about arithmetic on a table as much as about the sea. A fitted coefficient is a ratio of two sums over the table; a correlation coefficient is another; the size of a tail is a count of the events beyond a threshold that is itself a function of the table; the range of a quantity is its two extreme entries. Each is a finite computation, and because the table was published, each can be repeated by a reader with no access to the authors' environment. What a reader cannot do with the usual tools is say, of a printed "0.83", whether it is the rounding of the exact value, its truncation, or neither: floating-point arithmetic over sixteen thousand

terms does not settle the last digit, and the scripts' own procedures include randomised fitters whose output no re-run reproduces. This paper repeats every number the paper prints in exact rational arithmetic — every literal of the table read as the rational its digits denote, every sum an integer sum, every root and every π an enclosure — with the definitions the authors' own scripts use, and gives each printed number a verdict: the rounding of the exact value, its truncation, its integer parts, not the table's, or not decidable. A statement is called *certified* here, in the mathematical sense only, when it comes with a machine-checkable proof; for this table that is an exact rational computation from the pinned literals, or an enclosure whose two ends are such computations.

The contribution is therefore not a new measurement, a new scaling or a new test. It is (i) a verdict for each of the 25 numbers in the paper's text and each of the 140 cells of its Table 1, read against the public table; (ii) the identification, from the table alone, of the column that each printed quantity is — two of them turn out to be columns other than the ones their names suggest, which is stated as what the table shows; (iii) Duncan's laboratory band and ratio decided event by event, as the authors' scripts draw them; (iv) the stratification by sea state that the paper's conclusions name as requiring "a dedicated stratified analysis by H_s, T_p, or wave age, which is beyond the scope of this study", done per record with the paper's own fit and offered as the extension the dataset allows; and (v) the public, re-runnable record of all of it. Section 2 describes the dataset as published; Section 3 the arithmetic and the reading rule; Section 4 the results; Section 5 what a certificate adds to a dataset paper and what is not claimed.

2. The dataset

2.1. The measurements

The dataset is described by Guimarães et al. (2026a), and only what the present reading depends on is repeated. Twenty stereo-video records were acquired from the Marine Hydrophysical Institute platform off Katsiveli in the Black Sea, in about 30 m of water, with a pair of synchronised 5-megapixel cameras 11 m above the surface, on 9 days between 2013-09-22 and 2013-10-02, at 10 to 15 frames per second for 13 to 30 minutes each over a reconstructed area of 299 to 382 m^2; the reconstruction follows Benetazzo (2006); Benetazzo et al. (2012), and the data collection is described by Guimarães et al. (2020). As the paper's Table 1 prints them, the records span H_s from 0.35 to 1.88 m, T_p from 3.9 to 7.6 s and U_{10} from 6.1 to 18.0 $\mathrm{m\,s^{-1}}$, with 40 to 3,320 active breaking events per record. Active breaking

was detected by the pixel-intensity threshold of Mironov and Dulov (2008), with an event defined as foam connected in space and time (Kleiss and Melville, 2011); 22,116 events were detected automatically and 16,369 kept after manual inspection. For each event the contour of the foam patch at each frame gives its area A_b and its length L_b (half the perimeter of the contour), a minimum enclosing ellipse gives the across-wave and along-wave extents e_A and e_B, the three-dimensional reconstruction gives the vertical extent Δz_b, and the displacement of the barycentre between frames gives the speed c_b; the Duncan length L_{D81} and the inclination θ are built from Δz_b and e_B as Section 2.2 states.

2.2. The table as published

The Zenodo record (Guimarães et al., 2026b) is a single 8.9 GB archive of 145,057 entries, of which all but seven are per-event files and frames. The seven under `codes/` — the per-event table, a pandas pickle, and six analysis scripts — were extracted without downloading the archive, by reading its ZIP64 central directory over an HTTP range request and fetching each entry by its own range; each extracted file is pinned by its SHA-256 and the archive by the MD5 that Zenodo publishes (9349b0c9fc0bd2de2ba5aea50423232a). The pickle was converted once to a CSV in which every number is the shortest decimal that round-trips to the double the pickle holds, verified column by column against the pickle with a correctly rounded parser; that CSV (SHA-256 2e47b3af8725…) is the record every number below is computed from. It holds 16,369 rows and 46 columns; a row is one event, and a record's wind speed, peak frequency and the two columns discussed below are constant within the record (this is verified: no record carries two values).

The columns used, with the names the table gives them and the definitions the authors' loader script assigns (pinned, SHA-256 d763c15bd5e6…), are: `DT`, the duration Δt_b in seconds, which takes 29 distinct values because it is a multiple of the frame interval; `cm`, the mean speed c_b; `c0`, the initial speed; `cB`, the speed component along the minor axis, the paper's c_e; `Ab_max`, the largest area of the aerated region during the event, the paper's A_b; `Pb_max`, the largest perimeter, from which the scripts take $L_b = \texttt{Pb_max}/2$; `LA_max` and `LB_max`, the largest major and minor axes of the enclosing ellipse, the paper's e_A and e_B; `Ae_max`, the largest ellipse area; `Dz_max`, the largest vertical extent; `LD81` and `theta`, which the loader computes at the frame of the event's largest minor axis B as $L_{D81} = (\Delta z^2 + B^2)^{1/2}$ and $\theta = \tan^{-1}(\Delta z / L_{D81})$ in degrees, so that θ cannot exceed 45°; `Ab_Lmax`, the area at that same frame; `Ab_sum`, the area summed over the event's frames; `sv_fp`, the record's peak frequency; `wnd`, its wind speed; `svA` and `svT`, the

reconstructed area and the duration of the record. Two further columns are constant within each record: one named `Hs`, which takes 14 distinct values over the twenty records, between 0.1992 and 0.7520, each within one unit in the last place of the double of a multiple of 1/1024; and one named `sv_fp2`, whose twenty values lie between 0.3515 and 1.8834. In the loader, the list of column names and the assignment of values to them differ by one position across three columns, so that the column the public table names `sv_fp2` holds the quantity the loader reads as the significant wave height and the column it names `Hs` holds the quantity the loader names `sv_fb0`. Section 4.1 decides which of the two the paper's Table 1 prints.

2.3. The paper as pinned

The paper's numbers are read from the publisher's full-text XML, fetched on 2026-09-17 and pinned by its SHA-256 (baaa1076186d…), because the MathML in the XML carries digits that the HTML rendering drops. 25 printed numbers in the text and the 140 cells of Table 1 (twenty records, seven numeric columns) were transcribed with their location in the paper; the event counts of Table 1 sum to 16,369, as the abstract states. Two numbers printed in the paper are not about the table and are not decided: the 22,116 events detected before manual inspection, which are not in the table, and the $c_b \approx 0.8\,c$ of Banner et al. (2014a), which the paper quotes.

3. The decision

3.1. Exact arithmetic on the literals

Every numeric cell of the table is read as the rational number its decimal digits denote; the literal in the file is the number, never its floating-point image. Every quantity below is then a rational function of those literals computed in arbitrary-precision integer arithmetic: a ratio is a ratio, a comparison with a stated constant is an exact sign, a quantile is the k-th smallest value with $k = \lceil p\,n \rceil$ (a value that occurs in the table, never the average of two), and a least-squares slope through the origin, $k = \sum x_i y_i / \sum x_i^2$, is an exact rational of the literals once each series is multiplied up to a common denominator. Pearson's r is treated through its square, $r^2 = (n\sum xy - \sum x \sum y)^2 / [(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]$, which is exact; r itself is enclosed to twelve decimals by integer square roots, with its sign carried. Spearman's ρ is Pearson's r on average ranks (Spearman, 1904); the ranks are doubled so that tied values share an integer mean rank, and ρ^2 is again an exact rational. Only one quantity needs a transcendental constant, the breaking speed as a fraction of the peak phase speed

$c_p = g/(2\pi f_p)$, and there π enters as a certified enclosure (Moore et al., 2009; Rump, 2010), so that every quantile of c_b/c_p is an enclosure. The whole table is decided in 3.5 s.

3.2. The scripts' definitions

A printed number can be reproduced only under a definition, and the definitions are taken from the scripts the authors published, not chosen. A fit "$y = a\,c_b^2/g$" is `curve_fit` on the objective $a\,x^2/9.81$ with no intercept, that is, least squares through the origin on the regressor c_b^2/g with $g = 9.81$: $a = g\sum c_b^2 y / \sum c_b^4$, an exact rational. The r printed beside such a fit is `linregress`(c_b^2, y), the Pearson correlation of c_b^2 with y, and both it and the correlation of c_b with y are given. The self-similar subset of a ratio is the set of events with the ratio at or above its mean plus twice its sample standard deviation (`pandas`' `.std()`, `ddof` = 1) (McKinney, 2010). The mean of 16,369 ratios is a rational whose denominator is the least common multiple of their denominators and is useless as such; each ratio is instead enclosed by two integers at the scale 10^{30}, the sums are integer sums, and the mean, the variance and the standard deviation come out as enclosures 10^{-30} wide, decisive for any printed rounding. The threshold is therefore an enclosure, and an event is certainly at or above it, certainly below it, or — counted separately — inside the enclosure and undecided. The peak of a histogram is the mode bin of `numpy`'s histogram with the Freedman–Diaconis rule (Freedman and Diaconis, 1981; Harris et al., 2020): a bin width of $2\,\mathrm{IQR}\,n^{-1/3}$ with `numpy`'s linearly interpolated quartiles, $n^{1/3}$ enclosed by integer cube roots, and the bin count $\lceil \mathrm{range}/\mathrm{width} \rceil$ decided only when both ends of the enclosure give the same integer, after which the bins are exact. Two coefficients in the paper are read off a RANSAC regressor (Fischler and Bolles, 1981; Pedregosa et al., 2011), a randomised robust fitter whose output depends on its seed; they are not reproducible by any exact procedure and are reported as such, with the least-squares slope through the origin and the Pearson r — which the same script prints beside them — given instead.

3.3. The reading rule

A printed literal with d decimal places is read as the box of width 10^{-d} centred on it. The verdict is THE ROUNDING when the exact value lies strictly inside the box (it rounds, half away from zero, to the printed digits); THE TRUNCATION when it does not but the printed digits are the exact value's digits cut at the d-th place; A ROUNDING TIE when the exact value lies on the box's edge; THE INTEGER PARTS when a printed pair of integers is the integer parts of the two ends of a range under a named definition; NOT THE

TABLE'S when none of these holds under the definition the scripts use; and NOT DECIDABLE for a RANSAC coefficient. Where the exact value is an enclosure, both ends must give the same verdict, and otherwise the verdict is withheld at that width (this did not occur). The rule is one function, applied to the 25 numbers of the text and the 140 cells of Table 1 alike.

3.4. Controls and the record

A battery of 74 checks runs before any page or manuscript is written, with 12 red controls that must fire, among them: a non-numeric or comma-decimal literal is refused; a reversed line carries its sign; one literal of the table moved by 10^{-17} changes the exact r^2 of area with length; a printed 3 against an exact 1.376 is not the table's; a Table 1 count moved by one is refused; a printed 0.84 against an exact 0.8268 is refused; a value sitting on an enclosed threshold is undecided, never silently in or out; and a histogram whose range is an exact multiple of its bin width has its bin count refused rather than guessed. The battery then re-decides the whole table and the whole paper live and compares the result with the shipped record by digest. The record holds every exact value as a rational string beside its decimal reading, every enclosure as its two ends, and the digests of the 10 pinned files; any reader can recompute any entry from the CSV in any exact arithmetic.

4. Results

4.1. Table 1 read against the table

Table 1 reads the paper's Table 1 against the public table record by record. Every frame rate and every event count is exact; the durations and areas are the roundings of the record's `svT` and `svA` with one truncation; of the 140 cells, 128 are the rounding of the exact value, 7 its truncation, 3 a rounding tie and 2 neither. The printed H_s is the rounding or the truncation of the column named `sv_fp2` in 19 of the twenty records, and it is the rounding of the column named `Hs` in 0 of them: the significant wave height the paper prints is the column the public table names `sv_fp2`, which is what the loader's assignment of Section 2.2 shows. A reader of the public table who takes the column by its name reads a quantity between 0.1992 and 0.7520 instead, one that takes 14 distinct values over the twenty records; what that quantity is in the underlying per-event files, and what the two names refer to there, is for the authors to say. The paper's Table 1 is the correct column.

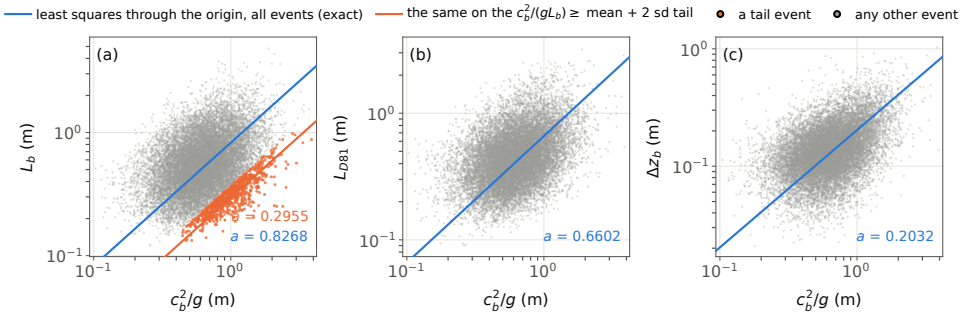


Figure 1: The three quadratic laws against the table: (a) L_b, (b) L_{D81} and (c) Δz_b against c_b^2/g for every event, on logarithmic axes, with the exact least-squares line through the origin ($a = 0.8268$, 0.6602, 0.2032). In (a) the 725 events at or beyond mean + 2 sd of $c_b^2/(gL_b)$ — the paper's self-similar subset — are drawn apart with their own fit ($a = 0.2955$). The points are the table's; the lines are the record's.

4.2. The fitted laws and the self-similar tail

The paper's central numbers survive exact re-derivation (Table 2, Fig. 1). The three quadratic laws of its Figure 3, least squares through the origin on c_b^2/g, have exact coefficients 0.8268, 0.6602 and 0.2032, of which the printed 0.83, 0.66 and 0.20 are the roundings, each inside its printed bootstrap interval. The self-similar fraction — every event at or beyond the mean plus two sample standard deviations of a ratio — is 725 events (4.43%) on $c_b^2/(gL_b)$, 678 (4.14%) on A_b/L_b^2 and 657 (4.01%) on A_b/L_{D81}^2, all inside the printed $4 \pm 2\%$, with the thresholds 2.6928, 0.6801 and 1.0877 and no event inside any threshold's enclosure. On the 725 tail events of $c_b^2/(gL_b)$ the fit is $gL_b/c_b^2 = 0.2955$ with $r(c_b^2, L_b) = 0.858$: the paper's 0.30 and 0.9. On the 15,644 remaining events the same fit is 0.8885 and $r = 0.446$; the printed 0.83 for that subset is the all-events coefficient, and the printed $r = 0.4$ is the rounding of the subset's 0.446 (and of the pooled $r(c_b, L_b) = 0.373$).

4.3. The correlations and the slopes of area on length

The correlations between geometric properties print as the roundings of their exact values: $r(A_b, L_b) = 0.9$ is 0.898, $r(A_b, L_{D81}) = 0.8$ is 0.808, $r(\Delta z_b, e_B) = 0.7$ is 0.679, $r(L_b, \Delta t_b) = 0.5$ is 0.475 and $r(\Delta z_b, \Delta t_b) = 0.4$ is 0.368. The printed $r(A_b, \Delta t_b) = 0.6$ is 0.444 with the largest area of the event and 0.613 with the area summed over its frames (the column `Ab_sum`); the table reproduces the printed value under the second definition. The sentence "Pearson $r < 0.45$ in all cases", about the alternative speeds, holds for 31 of the 32 speed–geometry pairs the table allows (four speeds against

Table 1: The paper's Table 1 (Guimarães et al., 2026a) read against the public table: the printed frame rate, duration, reconstructed area and event count N_b; the printed H_s beside the record's column named `sv_fp2` and its column named `Hs`; the printed T_p and U_{10} beside the record's $1/$`sv_fp` and `wnd`. A printed cell with no mark is the rounding of the exact value; ^{t} its truncation; ^{b} a rounding tie; ^{n} neither.

record (UTC)	Hz	min	m^2	N_b	H_s (m)	`sv_fp2`	`Hs`	T_p (s)	$1/$`sv_fp`	U_{10} ($\mathrm{m\,s^{-1}}$)	`wnd`
2013-09-22 12:00	12	30	335	127	0.54	0.541	0.551	4.7	4.72	6.1	6.06
2013-09-22 13:00	10	30	372	175	0.41	0.414	0.752	4.1	4.10	6.1	6.07
2013-09-23 10:40	12	30	372	1,074	0.46	0.460	0.551	4.0^{t}	4.07	9.6	9.59
2013-09-23 13:18	12	22	353^{t}	368	0.35	0.351	0.551	4.3	4.29	9.3	9.28
2013-09-23 13:55	12	30	354	438	0.40	0.397	0.586	4.5	4.51	9.9^{b}	9.95
2013-09-24 09:45	10	30	344	1,055	0.76	0.760	0.557	6.9	6.86	11.6	11.62
2013-09-25 10:00	12	25	299	1,882	0.91	0.906	0.504	4.5	4.51	14.9	14.87
2013-09-25 11:10	12	30	354	1,557	0.67	0.666	0.492	4.8	4.78	16.6	16.58
2013-09-25 12:15	12	30	372	3,320	0.65	0.646	0.480	5.7	5.66	15.2^{b}	15.25
2013-09-26 12:32	15	13	382	129	0.49^{t}	0.495	0.396	5.7	5.66	7.4	7.39
2013-09-27 07:35	12	30	316	461	1.16	1.158	0.516	7.6	7.61	14.0	14.00
2013-09-30 08:00	12	30	344	40	0.65^{n}	0.676	0.316	3.9	3.87	8.8^{b}	8.85
2013-09-30 10:20	12	30	299	386	0.66	0.664	0.328	4.2^{t}	4.25	8.7	8.71
2013-09-30 13:22	12	30	299	169	0.66	0.656	0.328	4.2^{t}	4.26	8.2	8.18
2013-10-01 06:00	12	30	299	1,161	1.57	1.568	0.258	7.0	6.96	15.7	15.70
2013-10-01 08:05	12	30	299	1,146	1.88	1.883	0.199	6.5^{t}	6.56	17.6	17.60
2013-10-01 08:50	12	30	299	1,188	1.72^{t}	1.725	0.258	7.1	7.09	18.0	17.99
2013-10-01 11:45	12	30	325	1,225	1.85	1.853	0.281	6.7^{n}	6.58	16.6	16.62
2013-10-02 05:45	12	30	299	195	0.97	0.974	0.328	6.1	6.12	7.6	7.58
2013-10-02 06:30	12	30	354	273	0.90	0.904	0.328	6.1	6.11	9.5	9.50

Table 2: The 25 numbers printed in the text of Guimarães et al. (2026a), the exact value under the scripts' definitions, and the verdict of Section 3.3. r beside a fit is Pearson's r of c_b^2 (or of the regressor) with the response, as `linregress` prints it.

printed quantity	printed	exact (the table, the scripts' definitions)	verdict
N	16,369	16,369	the rounding
θ, lower end	3	1.3763	not the table's
θ, upper end	71	42.9694	not the table's
θ range as $\tan^{-1}(\Delta z_b/e_B)$	3° to 71°	3.51° to 71.75°	the integer parts
mean A_b/L_{D81}^2	0.56	0.5656	the truncation
sd of A_b/L_{D81}^2	0.26	0.2610	the rounding
CV of A_b/L_{D81}^2 (%)	46	46.1500	the rounding
peak of A_b/L_{D81}^2	0.46	mode bin [0.498, 0.524)	not the table's
$L_b = a\,c_b^2/g$	0.83 ± 0.04	0.8268	the rounding
$L_{D81} = a\,c_b^2/g$	0.66 ± 0.04	0.6602	the rounding
$\Delta z_b = a\,c_b^2/g$	0.20 ± 0.02	0.2032	the rounding
r(speed, geometry)	< 0.45	0.4536 (speed, Δz_b)	not the table's
$r(A_b, L_b)$	0.9	0.8982	the rounding
$r(A_b, L_{D81})$	0.8	0.8079	the rounding
$r(\Delta z_b, e_B)$	0.7	0.6793	the rounding
$r(A_b, \Delta t_b)$	0.6	0.444 ($A_{b,\max}$); 0.613 ($A_{b,\mathrm{sum}}$)	not the table's
$r(L_b, \Delta t_b)$	0.5	0.4752	the rounding
$r(\Delta z_b, \Delta t_b)$	0.4	0.3676	the rounding
self-similar fraction	$4 \pm 2\%$	4.429%, 4.141%, 4.013%	the rounding
tail: gL_b/c_b^2	0.30, $r = 0.9$	0.2955; $r = 0.858$	the rounding
rest: gL_b/c_b^2	0.83, $r = 0.4$	0.8885; $r = 0.446$	not the table's
$A_b = k\,L_{D81}^2$	0.22	0.2696 (max. area); 0.2243 (area at max. length)	not the table's
$A_b = k\,L_b^2$	0.27	0.1726 (all); 0.7410 (tail)	not the table's
$e_A = k\,e_B$	2.37, $r = 0.8$	2.333 (least squares); $r = 0.760$	not decidable (RANSAC)
$A_e = k\,A_b$	2.5, $r = 0.9$	3.593 (least squares); $r = 0.793$	not decidable (RANSAC)

eight properties); the remaining pair, the mean speed against the vertical extent Δz_b of the paper's Figure 3c, is 0.4536. The two coefficients read off RANSAC fits, $e_A/e_B = 2.37$ and $A_e = 2.5\,A_b$, are not decidable; the least-squares slopes through the origin on the columns the script sets are 2.333 and 3.593, the $r = 0.8$ printed beside the first is the rounding of 0.760, and the $r = 0.9$ beside the second is the rounding of 0.862 on the summed columns (`Ae_sum`, `Ab_sum`) and not of 0.793 on the maxima.

The two printed slopes of area on length squared, "$A_b/L_{D81}^2 = 0.22$ and $A_b/L_b^2 = 0.27$", are read against every subset and both area columns. The slope of A_b on L_{D81}^2 through the origin is 0.2696 with the largest area of the event (`Ab_max`, the column the histogram, the mean and the tail use) and 0.2243 with the area at the frame of the largest minor axis (`Ab_Lmax`, the column the script's Figure 3d uses); the slope of A_b on L_b^2 is 0.1726 on all events, 0.1723 on the events outside the tail and 0.7410 on the tail. What the table shows is therefore that 0.2243 rounds to the printed 0.22, that 0.2696 rounds to the printed 0.27, and that no slope of A_b on L_b^2 rounds to either; the reading is left to the authors. The mean and spread of the aspect ratio are the table's: $A_b/L_{D81}^2 = 0.56 \pm 0.26$ with a coefficient of variation of 46% is, exactly, 0.5656 (printed as its truncation) $\pm$ 0.2610 (the sample standard deviation; the population value is 0.2610) with 46.15%. The histogram of that ratio under the Freedman–Diaconis rule on the table has 78 bins of width 0.0259, a single mode bin $[0.498, 0.524)$ with 748 events, and no tie; the printed peak of 0.46 lies two bins to its left. The paper's Figure 2d describes its ratio as "calculated at the maximum length", and the table holds the area at that frame as a separate column; the mode of that column's ratio was not decided here.

4.4. The range of the inclination

"The range of θ observed here (3° to 71°)" is not the range of the column the table names `theta`, which the histogram script plots and which runs from 1.38° to 42.97° with no event beyond 45°, as the loader's definition $\theta = \tan^{-1}(\Delta z/L_{D81})$ with $L_{D81} = (\Delta z^2 + B^2)^{1/2}$ requires. It is, to the integer part, the range of a different angle in the same table: $\tan^{-1}(\Delta z_{b,\max}/e_{B,\max})$, whose tangent runs from 0.0614 to 3.0331 over the events, that is, from 3.51° to 71.75°, decided through enclosed tangents of 3°, 4°, 71° and 72°. The paper takes Δz_b as equivalent to $L_{D81}\sin\theta$; the table holds both angles, and which one the sentence intends is for the authors to say. Duncan's band of 10° to 14.7°, drawn by the authors' script on the column named `theta`, is decided on that column in the next section.

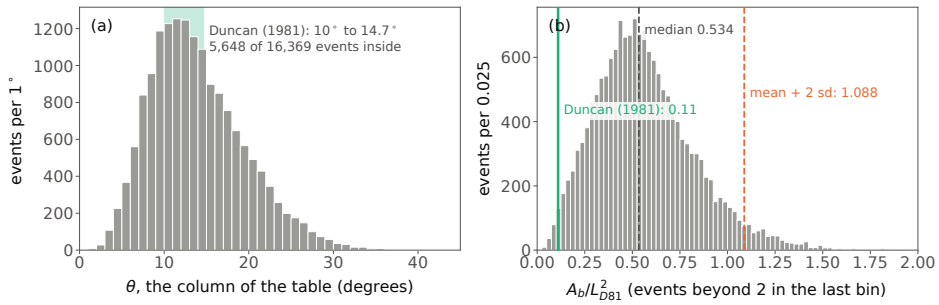


Figure 2: Duncan's references as the authors' script draws them, on the table: (a) the inclination θ of the column named `theta`, in 1° bins, with the band 10° to 14.7° that holds 5,648 events; (b) the aspect ratio A_b/L^2_{D81} in bins of 0.025, with the laboratory value 0.11, the median 0.5343 and the tail threshold mean + 2 sd of the paper's self-similarity test. Bin membership is drawn in float; every count stated in the text is exact.

4.5. Duncan's geometry as the record treats it

The authors' histogram script draws two laboratory references (Duncan, 1981): the band 10° to 14.7° on the inclination and the line 0.11 on the aspect ratio A_b/L^2_{D81}. Both are decided event by event with no tolerance, by an exact comparison of the literal with the stated constants (Fig. 2). Of the 16,369 events, 5,648 (34.5%) have their inclination strictly inside the band, 4,280 (26.1%) are flatter, 6,441 (39.3%) are steeper and 0 lie on an edge. The aspect ratio lies above 0.11 at 16,194 events and below it at 175, with 263 events within a quarter of it and 550 within half; its median, 0.5343, is 4.86 times the laboratory value, and the ratio runs from 0.032 to 2.053 over the table. One caveat belongs to the comparison itself and is the authors' as much as this paper's: Duncan's 0.11 relates the cross-sectional area of the breaking region to its length, whereas the area measured from above a natural sea is the plan view of the whitecap. The script draws the line on that histogram, the paper prints the comparison as 0.56 ± 0.26 against 0.11 ± 0.01, and the comparison is decided here as drawn.

4.6. Three ratios a self-similar breaker would hold constant

If breaking waves were geometrically similar, a length made from the speed, c_b^2/g, would be a fixed multiple of the measured length and the area a fixed multiple of the length squared: each of the paper's three ratios would be a constant up to measurement noise. Table 3 gives their order statistics, each the k-th smallest exact ratio. The middle half of the events spans a factor of about two in every ratio (q_{75}/q_{25} = 2.06, 1.87, 1.87) and the

Table 3: The three self-similarity ratios as order statistics of the exact values (q_p is the $\lceil p\,n \rceil$-th smallest), their spread factors, and the paper's tail: the threshold mean + 2 sd (sample sd, enclosed; the lower end is shown) and the exact count and share of events at or beyond it. No event lies inside a threshold's enclosure.

ratio	q_{05}	q_{25}	median	q_{75}	q_{95}	q_{75}/q_{25}	q_{95}/q_{05}	mean + 2 sd	tail	share
$c_b^2/(gL_b)$	0.452	0.780	1.133	1.607	2.606	2.06	5.77	2.693	725	4.43%
A_b/L_b^2	0.156	0.245	0.338	0.459	0.664	1.87	4.26	0.680	678	4.14%
A_b/L_{D81}^2	0.191	0.381	0.534	0.714	1.040	1.87	5.43	1.088	657	4.01%

Table 4: Rank and linear correlations between the breaking speed and the geometric properties, and between geometric properties, over the 16,369 events: Spearman's ρ (an exact rational, its root read to three decimals) and Pearson's r on the literals.

pair	Spearman ρ	Pearson r
c_b and L_b	0.360	0.373
c_b and A_b	0.375	0.355
c_b and Δz_b	0.410	0.454
c_b and Δt_b	0.150	0.147
c_b and L_{D81}	0.361	0.367
c_b and θ	0.097	0.109
A_b and L_b	0.902	0.898
Δz_b and L_b	0.649	0.678
L_b and Δt_b	0.397	0.475

central ninety per cent a factor of four to six (q_{95}/q_{05} = 5.77, 4.26, 5.43). The paper's test (Patel and Schoenberg, 2011; Clauset et al., 2009) marks the tail beyond the mean plus two standard deviations and calls the events in it self-similar; the spread factors are the same finding in a form that depends on no distribution. The rank correlations the paper's second finding rests on are in Table 4: the breaking speed ranks with any geometric property at $\rho \leq 0.410$ (with Δz_b; $\rho^2 = 0.168$), against 0.902 for area with length. The breaking speed is a fraction 0.284 of the peak phase speed at the median event (the central ninety per cent from 0.193 to 0.412, each an enclosure through π), and 127 events are faster than half of it.

4.7. The stratification by sea state

The paper's abstract states that "geometric scaling is not constant but varies with the sea state"; its discussion notes that the fitted coefficients vary from record to record and that a dedicated stratified analysis by H_s, T_p or wave age would be required to test the hypothesis, and places that analysis beyond its scope. The table allows it at the level of the record: each of the twenty records is one sea state, with H_s, T_p and U_{10} constant

within it, and the paper's own fit — least squares through the origin on c_b^2/g — can be made on each record's events alone. Table 5 gives, per record, the fitted coefficient gL_b/c_b^2, the correlation $r(c_b^2, L_b)$, the share of events inside Duncan's band, the mean aspect ratio and the share of events in the $c_b^2/(gL_b)$ tail; Fig. 3 draws the coefficient against the four sea-state variables, and Table 6 gives the rank correlations across the twenty records, each ρ^2 an exact rational on twenty ranks (the wave age $c_p/U_{10} = g/(2\pi f_p U_{10})$ needs no π for a rank: its order is the order of $1/(f_p U_{10})$).

The coefficient runs from 0.665 to 1.106 — a factor of 1.66 — around the pooled 0.8268, and the 4 records with H_s above 1.5 m, all of 1 October 2013 with winds of 15.7 to 18.0 $\mathrm{m\,s^{-1}}$, carry the four lowest values, 0.665, 0.704, 0.736, 0.740: in the roughest seas a breaker of a given speed is shorter. Across the twenty records the coefficient ranks with H_s at $\rho = -0.493$ and with T_p at -0.492, with U_{10} at -0.365, and with wave age at -0.026. The within-record correlation $r(c_b^2, L_b)$ runs from 0.099 to 0.420 and never approaches the 0.858 of the pooled tail; the share of events inside Duncan's band runs from 23.1% to 43.3% and ranks with nothing (with H_s at -0.286, with U_{10} at 0.021); the mean aspect ratio runs from 0.431 to 0.724 and ranks with H_s at -0.220. The share of a record's events in the self-similar tail runs from 0.0% to 11.6% and is concentrated in the four roughest records (10.5%, 11.6%, 9.8%, 9.2% of their events against 4.4% overall), which is the direction of the paper's observation that the self-similar events are the smaller and faster ones relative to the sea they occur in. Twenty points do not make a law, and the records are neither independent nor equal in size (3,320 events in the largest, 40 in the smallest); what they make is the sentence the paper's abstract states, with a number under it and the variable named — at the level of the record, the dependence is on the wave height and the period, and not on the wave age. A stratification by wave age alone, which the paper's discussion names first, would not show it on this table.

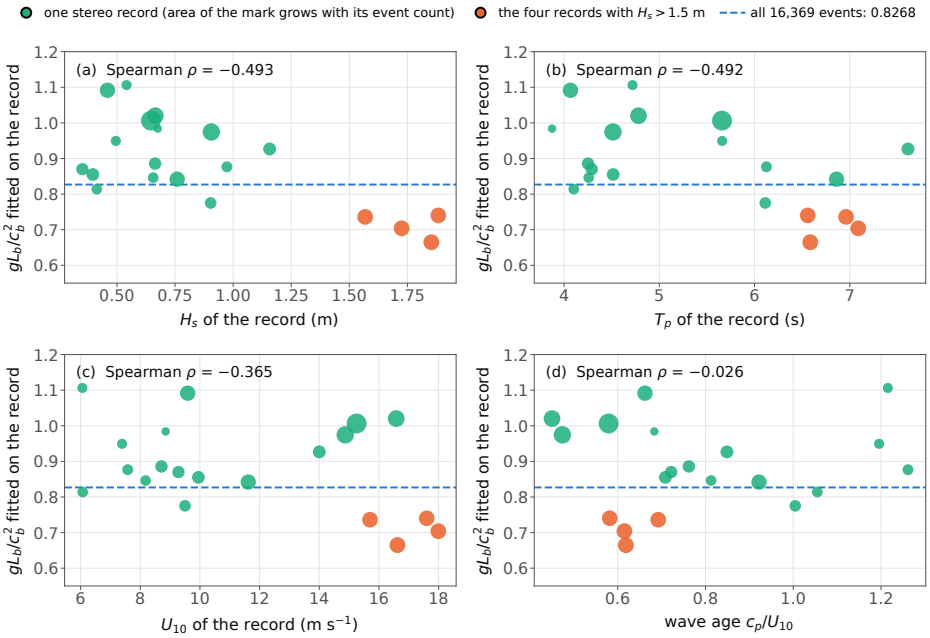


Figure 3: The coefficient of $L_b = a\,c_b^2/g$ fitted separately on each record's events, the same least squares through the origin the paper uses on all 16,369, against the record's (a) H_s, (b) T_p, (c) U_{10} and (d) wave age c_p/U_{10}, with the pooled coefficient as a dashed line and the four records with $H_s > 1.5$ m drawn apart. Spearman's ρ on the twenty ranks is from the record; the wave age is drawn from the record's T_p and U_{10}.

Table 5: The stratification by record: the number of events, the record's H_s, T_p and U_{10} (the columns `sv_fp2`, $1/$`sv_fp` and `wnd`), the coefficient gL_b/c_b^2 of the paper's fit made on the record's events alone, the correlation $r(c_b^2, L_b)$ within the record, the share of the record's events inside Duncan's band, the mean of A_b/L_{D81}^2 (enclosed; the lower end is shown) and the share of the record's events in the pooled $c_b^2/(gL_b)$ tail.

record (UTC)	events	H_s (m)	T_p (s)	U_{10} (m s^{-1})	gL_b/c_b^2	$r(c_b^2, L_b)$	in band (%)	mean A_b/L_{D81}^2	in tail (%)
2013-09-22 12:00	127	0.54	4.7	6.1	1.106	0.099	43.3	0.481	3.9
2013-09-22 13:00	175	0.41	4.1	6.1	0.814	0.190	38.9	0.698	1.7
2013-09-23 10:40	1,074	0.46	4.1	9.6	1.092	0.241	36.1	0.606	0.5
2013-09-23 13:18	368	0.35	4.3	9.3	0.870	0.246	36.4	0.641	1.9
2013-09-23 13:55	438	0.40	4.5	9.9	0.855	0.242	31.5	0.655	2.7
2013-09-24 09:45	1,055	0.76	6.9	11.6	0.842	0.194	41.9	0.597	6.5
2013-09-25 10:00	1,882	0.91	4.5	14.9	0.975	0.302	32.7	0.613	1.9
2013-09-25 11:10	1,557	0.67	4.8	16.6	1.020	0.301	35.1	0.513	0.7
2013-09-25 12:15	3,320	0.65	5.7	15.3	1.006	0.401	36.2	0.537	1.1
2013-09-26 12:32	129	0.49	5.7	7.4	0.949	0.325	25.6	0.564	1.6
2013-09-27 07:35	461	1.16	7.6	14.0	0.927	0.371	31.4	0.724	2.8
2013-09-30 08:00	40	0.68	3.9	8.8	0.984	0.404	32.5	0.431	0.0
2013-09-30 10:20	386	0.66	4.3	8.7	0.886	0.405	33.4	0.528	3.4
2013-09-30 13:22	169	0.66	4.3	8.2	0.846	0.377	23.1	0.511	4.1
2013-10-01 06:00	1,161	1.57	7.0	15.7	0.736	0.366	33.3	0.556	9.8
2013-10-01 08:05	1,146	1.88	6.6	17.6	0.740	0.388	31.7	0.547	9.2
2013-10-01 08:50	1,188	1.73	7.1	18.0	0.704	0.372	32.8	0.548	11.6
2013-10-01 11:45	1,225	1.85	6.6	16.6	0.665	0.350	33.8	0.536	10.5
2013-10-02 05:45	195	0.97	6.1	7.6	0.877	0.420	30.8	0.492	3.6
2013-10-02 06:30	273	0.90	6.1	9.5	0.775	0.392	31.9	0.534	4.0

Table 6: Spearman's ρ across the twenty records between a per-record quantity and the record's sea state; each ρ^2 is an exact rational on twenty ranks, read to three decimals with its sign.

per-record quantity	H_s	T_p	U_{10}	c_p/U_{10}
coefficient gL_b/c_b^2	−0.493	−0.492	−0.365	−0.026
mean A_b/L_{D81}^2	−0.220	0.123	0.171	−0.044
median $c_b^2/(gL_b)$	0.391	0.386	0.361	−0.036
share of events in 10°–14.7°	−0.286	−0.170	0.021	−0.114

5. Discussion

What a certificate adds to a dataset paper. A dataset paper is already the most reproducible kind of paper: its table is public and its numbers are functions of it. What the exact re-run adds is the last digit and the definition. The printed digits become a box, the exact value either rounds into the box or does not, and when it does not the record says which column, which subset or which definition does reproduce the digits — the summed area for $r(A_b, \Delta t_b)$, the area at the frame of maximum length for the slope 0.22, the angle $\tan^{-1}(\Delta z_b / e_B)$ for the printed range of θ, the column named `sv_fp2` for the printed H_s. None of these is visible in the printed digits, and all of them are visible in the record. Refusal is a result of the same kind: the two RANSAC coefficients are recorded as not decidable, with the deterministic slope and the r beside them, rather than re-run with a seed and called reproduced; a histogram's bin count is refused when the enclosure of $n^{1/3}$ straddles an integer rather than guessed. For a reader of the public table the column identity is the practical part: the column named `Hs` is not the significant wave height of the records, and the column named `theta` is an angle that cannot exceed 45°, so that the paper's Table 1 and its printed range of θ are the quantities to use. The grammar — a verdict per printed number, with the exact value and the definition under it, and a record that can be re-run without the authors' code — is what is proposed, and it is independent of this dataset.

What is not claimed. Nothing about the ocean is decided here, only about the table and about the paper's arithmetic on it. The table is the authors' measurement, with whatever error the stereo reconstruction and the white-cap detection carry; the record describes neither, and a stereo rig's geometry alone can move a length by an amount that the companion instrument of the same project bounds (Toledo, 2026). Duncan's band and ratio are taken as the authors' scripts state them, and the plan-view caveat of the aspect-ratio comparison is real. The 22,116 events detected before manual inspection are not in the table and cannot be checked. The Kolmogorov–Smirnov p-values, the Akaike ranking of the distributions of a scientific library (Virtanen et al., 2020), the Pareto and log-normal fits, the kernel densities, the bootstrap confidence intervals on the fitted coefficients and the maximum-likelihood cross-check of the tail are floating-point procedures about a distribution and are not reproduced; a RANSAC coefficient is a random variable and is reproducible by nothing. The stratification is by record, and twenty records of unequal size from 9 days at one site are a description of this table, not a

law; it is offered as the extension the dataset allows, in the form the paper's discussion asks for, for the authors to take further with the wave ages and wind directions the records carry.

Cost. The whole table — 16,369 events, every ratio, quantile, correlation and fit, the 25 printed numbers, the 140 cells and the twenty records — is decided in 3.5 s of integer arithmetic on a laptop; the battery of 74 checks takes a few seconds more. The arithmetic is arbitrary-precision rational arithmetic with no floating point anywhere except the one enclosure of π, and the cost of exactness at this scale is nil.

6. Conclusions

Every number printed in a dataset paper on the geometry of 16,369 breaking waves was re-decided in exact arithmetic from the paper's own public table with the definitions of the paper's own scripts. 15 of the 25 numbers in the text and 128 of the 140 cells of its Table 1 are the rounding, the truncation or the integer parts of the exact value; the three quadratic laws, the correlations between geometric properties, the $4 \pm 2\%$ self-similar tail and the fit on it reproduce to the printed digit. Two printed quantities are shown to be other columns of the public table than the ones their names suggest, and the record states which; two are coefficients of a randomised fitter and are recorded as not decidable. Duncan's laboratory geometry is decided event by event, with 34.5% of the events inside his band of inclination and the aspect ratio's median 4.86 times his value. The stratification by sea state that the paper leaves for a dedicated analysis is done per record on the same table with the same fit: the coefficient of L_b on c_b^2/g spans a factor of 1.66 across the twenty records and ranks with the wave height and the period, not with the wave age. What is proposed to the authors of such datasets is not a different number but a different object to accompany each printed one: a verdict, the exact value under it, and a record that any reader can re-run.

Declaration of competing interest

The author declares no competing financial or personal interests. The dataset paper is cited as the reference work; its authors are being consulted on this manuscript.

Data availability

The record, the instrument and the figure and table sources are public in the cert-machine repository (`https://github.com/carlostoledo1891/cert-machine`, commit 9349d8bad505): the ledger `certs/breaking-ledger.json` (generated 2026-09-17), the pinned corpus under `corpus/blacksea-breaking` (the per-event table as a CSV, the seven files of the Zenodo record with their archive offsets and digests, the transcription of the paper's printed numbers, and the digest of the paper's full-text XML), the instrument under `instruments/breaking` (MIT) and the number and figure builders under `tools/paper-numbers`. The report with every event re-decided at each build is at `https://carlostoledo.co/reports/breaking-geometry.html`. The dataset is Guimarães et al. (2026b) (CC BY 4.0); the paper is Guimarães et al. (2026a).

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used large language models (Claude, Anthropic) to draft code, exposition and this manuscript in a certified-verification setting: constructions and prose were proposed with the help of the models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for a record to be accepted. After using the tool the author reviewed and edited the content and takes full responsibility for the content of the publication. The records, not the models, are the evidence offered; they are public and re-runnable.

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